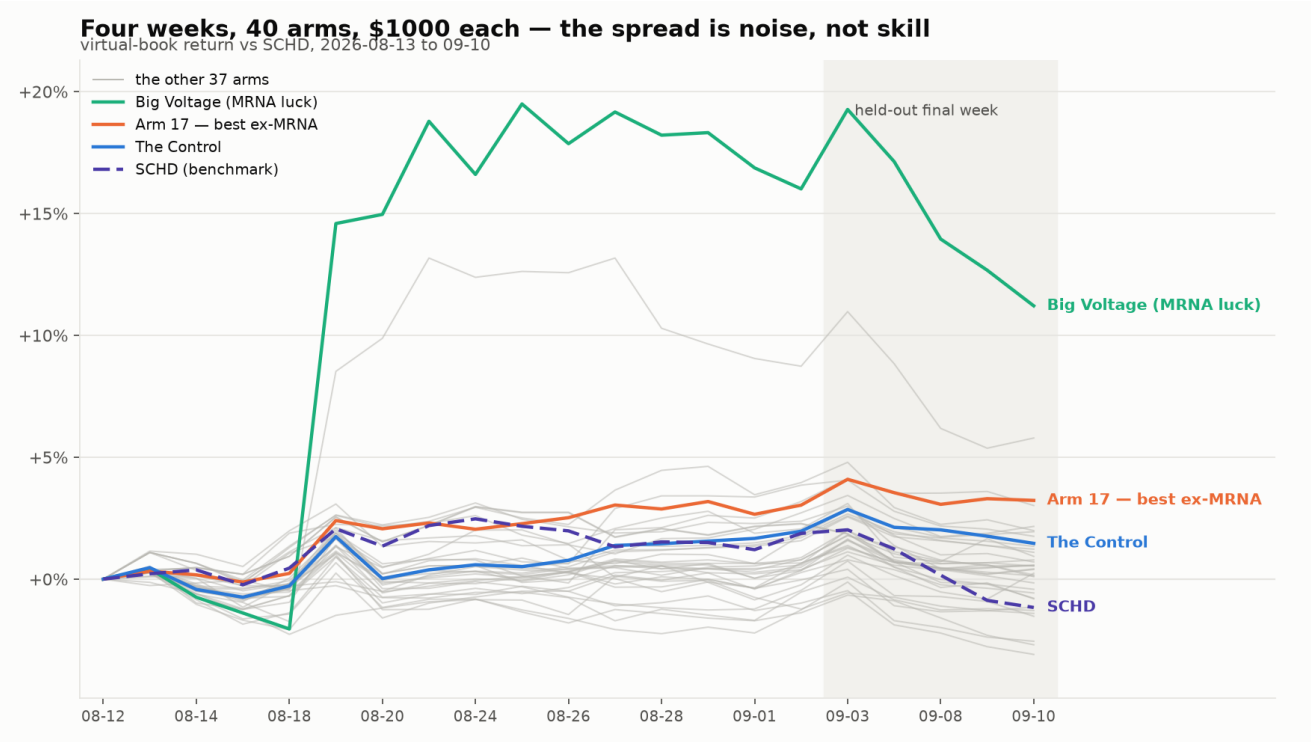


Stonkbot Arena — final analysis

Window 2026-08-13 → 2026-09-10 (20 trading days). 40 arms, \$1000 virtual book each, shared live market data. Experiment container stopped 2026-09-11 00:03 EDT. Benchmark SCHD 34.37 → 33.97 = **-1.16%**, split +1.88% (weeks 1-3) / -2.98% (final week). 8,269 closed trades.



Four weeks, 40 arms, \$1000 each. The spread is noise.

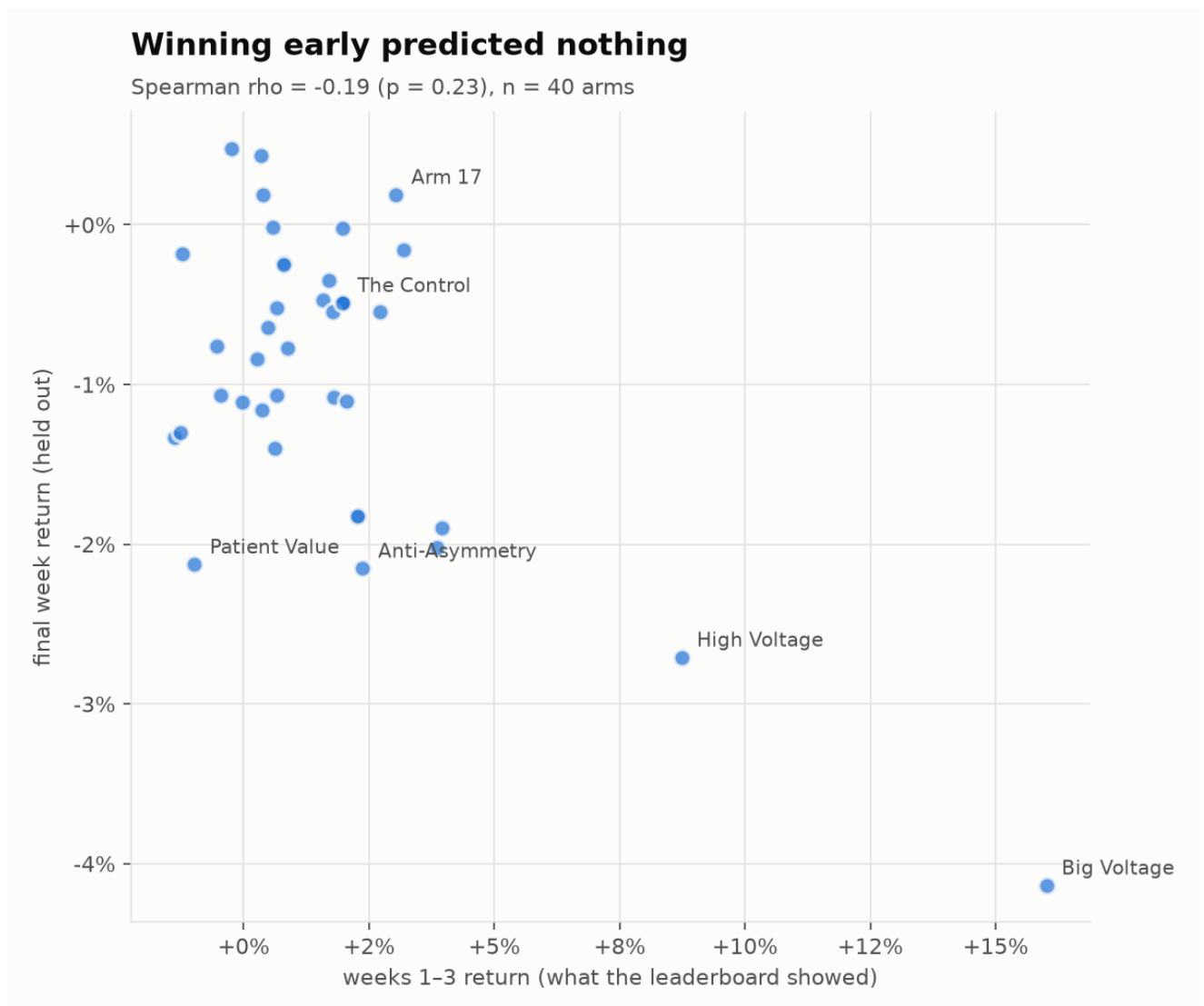
Verdict against the pre-registered rule

The rule was: rank on expectancy + profit factor, leader must beat **SCHD and the Control** by more than a bootstrap band, final week held out, <30 trades = inconclusive, and if nothing clears → **hold SCHD**.

TEST	RESULT
arms clearing SCHD, 95% band on daily excess	0 / 40
arms clearing the Control, 95% band on daily excess	0 / 40
arms clearing the Control on per-trade expectancy	0 / 40
arms beating SCHD on raw return, MRNA excluded	32 / 40
IS → OOS rank stability (expectancy)	rho +0.02 , p = 0.91
IS → OOS rank stability (return)	rho -0.19 , p = 0.23

Nothing clears. The pre-registered answer is to hold SCHD.

The rank-stability number is the real finding: how an arm placed over weeks 1-3 carried no information about week 4. The leaderboard was measuring the market, not the strategy.



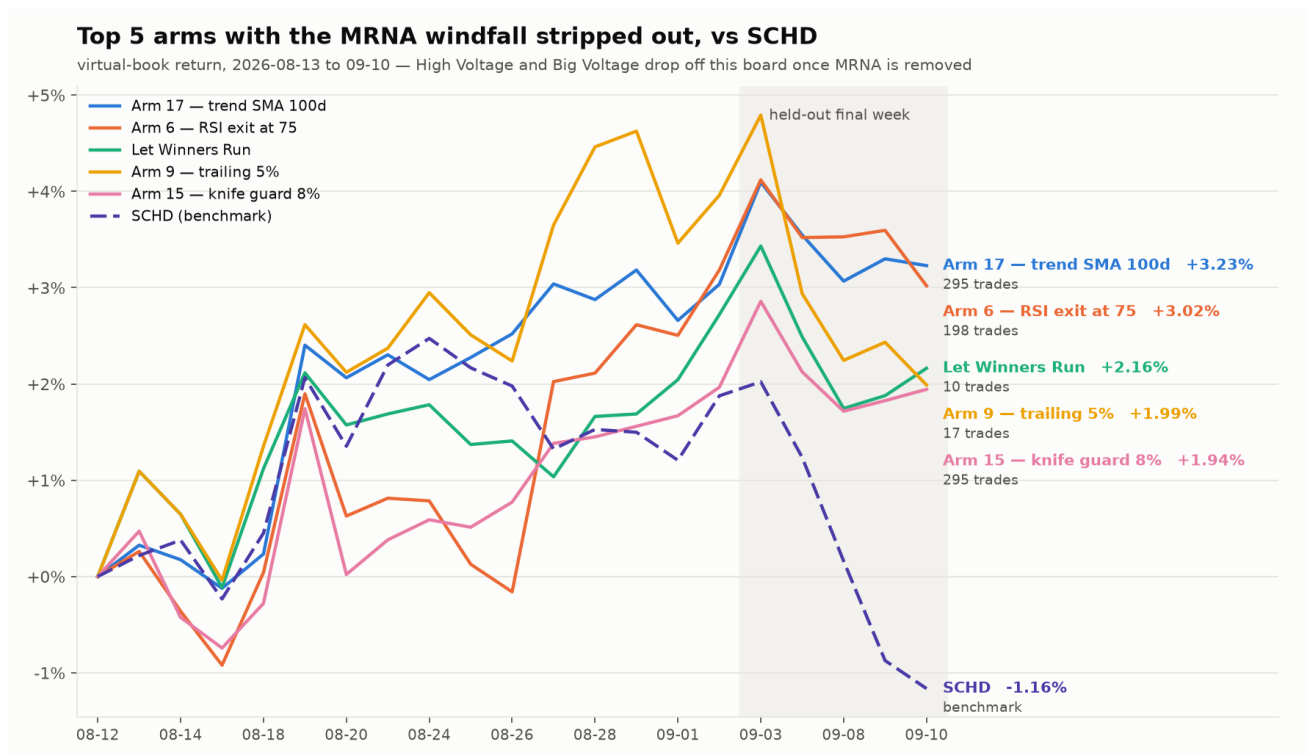
Weeks 1-3 placement said nothing about week 4.

The MRNA problem, confirmed

The two arms that topped the raw board are the two worst arms once MRNA is removed.

ARM	RAW RETURN	MRNA P&L	RETURN EX-MRNA
37 Big Voltage	+11.21%	\$172	-6.04%
28 High Voltage	+5.79%	\$106	-4.82%

Both also gave the gain back: Big Voltage peaked near +19% on 09-03 and finished +11%. 71 MRNA trades out of 8,269 decided the entire raw ranking.



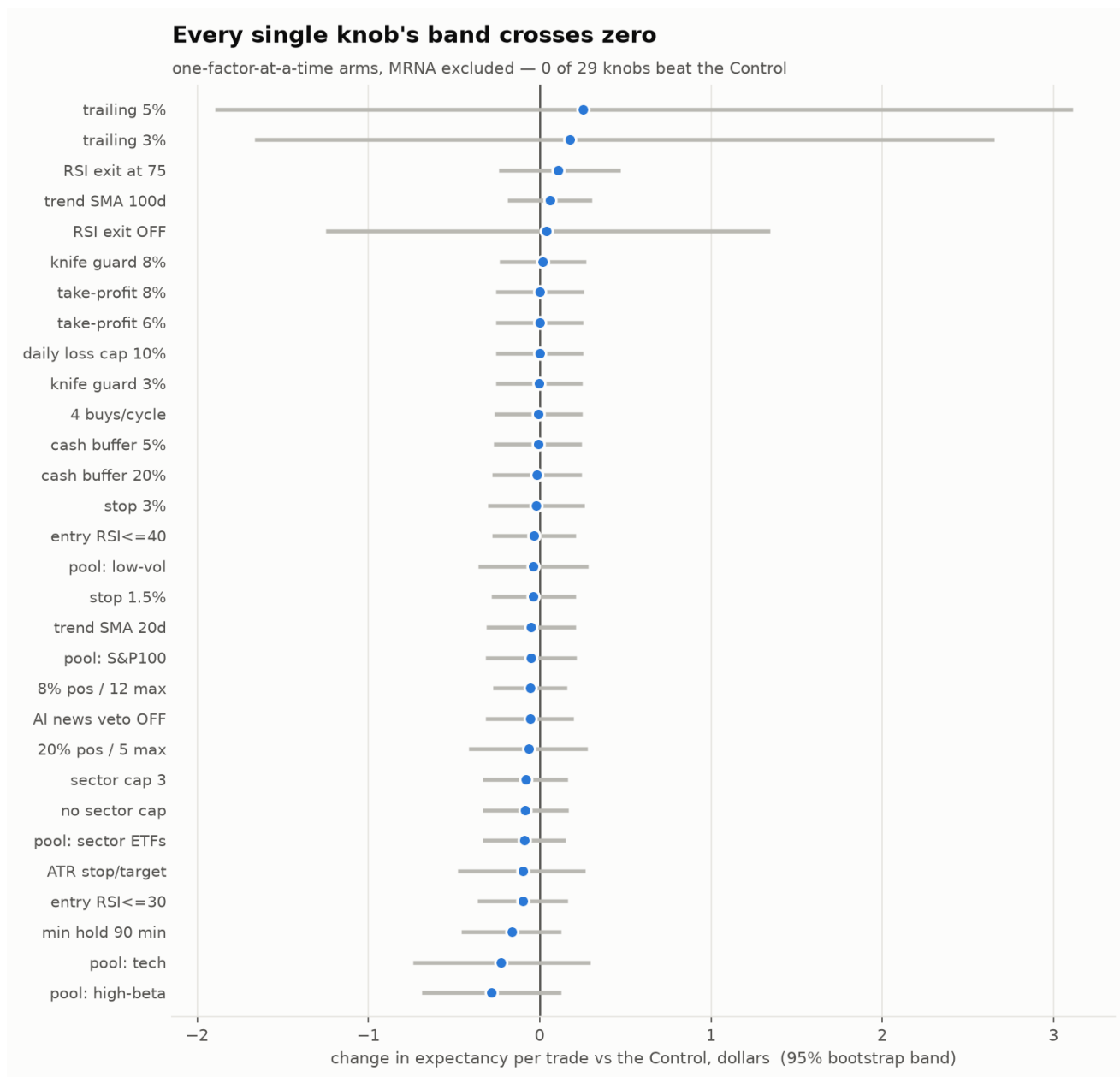
Top five once the MRNA windfall is stripped out.

Best and worst knobs (all statistically indistinguishable from the Control)

Point estimates only — every 95% band crosses zero.

Knob	Return vs Control
trend SMA 100d (arm 17)	+1.76 pp
RSI momentum exit at 75 instead of 65 (arm 6)	+1.55 pp
trailing 5%, no TP/RSI (arm 9)	+0.52 pp
...	
AI news veto OFF (arm 18)	-1.62 pp
min hold 90 min (arm 11)	-4.16 pp
pool: high-beta (arm 28)	-6.29 pp

The news veto being worth +1.6 pp when left on is the one directional result worth keeping, and it is not significant either.



Every knob's 95% band crosses zero.

Three arms produced no information

Arms 4 (take-profit 6%), 5 (take-profit 8%) and 26 (daily loss cap 10%) are **bit-for-bit identical to the Control** — same 292 trades, same \$21.63 realized, same final equity to ten decimals. Cause: with the RSI-65 momentum exit on, the exit fires in the same 5-minute cycle the take-profit would have, at the same last-close fill, so the take-profit level is dead code. And the 6% daily loss cap never bound in 20 days. A future design should test the take-profit only in arms that also disable the momentum exit.

The live bot's current config ranks badly

Arm 32 Anti-Asymmetry was promoted to the live bot on 2026-08-23 off a one-week read. Over the full window it returned +0.17% against the Control's +1.47%, with expectancy - **\$0.44/trade**, profit factor 0.75, and the final week at **-2.15%**. Only 23 closed trades, so it is formally inconclusive (<30) — but nothing in the data supports keeping it over the Control.

Structure of the edge

EXIT REASON	TRADES	TOTAL P&L	AVG
RSI momentum	6,628	+\$3,650	+\$0.55
stop-loss	1,442	-\$4,201	-\$2.91
take-profit	85	+\$708	+\$8.33
trailing	84	+\$415	+\$4.93
ATR	30	-\$178	-\$5.93

The original diagnosis holds: the Control wins 84.6% of trades, average win \$0.60, average loss -\$2.79 (ratio 0.21). The arms that fix the asymmetry (trailing stops, no RSI exit) do flip the ratio to 2.1-2.9, but they close so few trades (10-40 over four weeks) that their advantage never becomes measurable. Fixing the asymmetry and keeping enough trades to prove it are in direct conflict at this holding period.

Caveats

- 20 trading days is short. The bands are wide because the sample is small, not because the method is weak.
- Zero slippage, zero commission, fills at last close. Real execution can only be worse.
- The final week was a market drawdown (SCHD -2.98%); the arms lost less than SCHD mostly because they sat ~90% invested in a different (mega-cap) basket, not because of any knob.